PTRR FRAMEWORK

A Pre-Registered Study Design

Investigating Expertise as a Cross-Level Moderator of Cognitive Bias in AI-Assisted Vulnerability Research

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Status	Pre-Registered — Data Collection Pending Platform Deployment

1. Abstract

Transparency Note: *The initial evidence in this paper comes from a self-study conducted by the researcher himself. During that study, AI was used only as a temporary feedback tool — not as a formal evaluation instrument. Any final conclusions will depend on formal human scoring after participant recruitment is complete.*

This study looks at what we call the "Dual-Edge Effect" of AI in cybersecurity. In simple terms: AI helps experienced researchers do better work, but it may cause less experienced researchers to stop thinking for themselves. We built the PTRR Framework (Prompts, Time, Results, Rubric) to measure this difference in a real research setting. A key idea in this study is that a researcher's level of expertise changes how much AI helps or harms their work. This is called a moderating effect — expertise does not come from using AI, it shapes how AI is used.

2. Theoretical Anchoring

This study draws on two established frameworks. Dual Process Theory (Kahneman, 2011) explains Automation Bias as a System 1 failure: novices tend to accept AI output uncritically rather than engaging deliberate verification. Cognitive Load Theory (Sweller, 1988) explains why this effect is expertise-dependent — AI may eliminate the productive cognitive struggle necessary for novice skill development, while simultaneously freeing expert working memory for deeper analysis.

Together, these frameworks predict that expertise moderates the relationship between AI assistance and research quality, such that the same level of AI Interaction Quality will produce systematically different cognitive and output outcomes depending on the practitioner's skill level. This is a moderation claim, not a mediation claim: expertise does not arise from AI interaction, but rather determines how AI interaction is processed and utilized.

3. Conceptual Diagram

The figure below illustrates the study's analytical model, including the independent variable, dependent variable, cross-level moderator, additional moderator, and within-subject factor.

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┌──────────────────────────┐
│  AI Interaction Quality  │    ← Independent Variable
│   (PTRR Process Score)   │
└──────────────────────────┘
              │
              ▼
┌──────────────────────────┐
│    Research Output       │    ← Dependent Variable
│  (PTRR Results Score)    │
└──────────────────────────┘

Moderated by:
├─ Expertise Tier (Cross-Level Moderator)
├─ Tool Integration Intensity Score / TIIS
└─ Phase A vs. Phase B (Within-Subject Factor)
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4. Research Questions and Pre-Registered Hypotheses

Research Questions

RQ1: How does expertise level moderate the relationship between AI Interaction Quality and Research Output?

RQ2: To what extent do novices exhibit Automation Bias when AI provides false negatives or surface-level findings?

RQ3: Does the integration of traditional security tools alongside AI mitigate or amplify cognitive dependency? *(Exploratory — no directional hypothesis.)*

Pre-Registered Hypotheses

H1	Expert participants (Tier 3) will exhibit significantly lower Automation Bias Index scores during Phase B (AI-enabled) compared to novice participants (Tier 1), reflecting greater critical engagement with AI output rather than passive acceptance.
H2	Cognitive Struggle Index will positively predict Research Output Score in Tier 3 participants, but negatively predict Research Output Score in Tier 1 participants, consistent with the differential effect of desirable difficulty across expertise levels.
H3	Tool Integration Intensity Score will moderate the AI Interaction Quality – Research Output relationship such that higher TIIS attenuates Automation Bias. This moderation effect is expected to be stronger in Tier 2 and Tier 3 than in Tier 1.

5. Methodology & Environment

5.1 The Sandbox Platform

The study will be conducted on a custom-built web platform under a naturalistic observation design. No restrictive System Prompt is imposed on the AI, preserving ecological validity and avoiding artificial constraints on participant behavior.

- Interface: Participants interact with an LLM through a standard chat interface.
- Tool Usage: Participants are encouraged to use external tools (Burp Suite, Nmap, etc.) and feed the resulting data to the AI for analysis.
- Passive Monitoring: The platform silently records keystroke dynamics, data ingestion patterns, interaction linearity, and temporal sequencing of tool and AI usage.

5.2 Within-Subject Control Condition

Each session is structured in two phases to isolate the causal role of AI assistance and enable intra-participant comparison:

- **Phase A (30 minutes):** AI access is muted. Participants conduct vulnerability research using only traditional tools and manual reasoning, establishing a per-participant behavioral and cognitive baseline.
- **Phase B (30 minutes):** Full AI access is enabled. Participants may interact with the LLM freely alongside their toolset.

Comparing Phase A and Phase B within the same participant controls for individual differences in skill and motivation, and allows direct measurement of the cognitive shift induced by AI availability. PTRR process scores, ABI, and CSI are computed separately for each phase.

Pilot Study Note: A pilot study involving 5–8 participants will be conducted prior to full data collection to evaluate whether the 30-minute phase duration sufficiently captures investigative complexity — particularly for Tier 3 participants whose non-linear workflows may require longer observation windows to manifest. Phase duration will be adjusted before main data collection if pilot findings indicate a systematic floor effect in expert behavioral patterns.

5.3 Participant Tiers

Target: 30–50 (could increase it) participants, classified into three tiers. Sample size is informed by comparable HCI studies reporting medium-to-large effect sizes (Cohen's $d \geq 0.6$), for which $n = 10$–17 per group provides adequate power at $\alpha = 0.05$, $\beta = 0.80$.

Tier	Criteria	Profile
Tier 1 — Junior	< 20 reports, no verified	High reliance on automated scanners

	bounties	
Tier 2 — Mid	20–50 reports, verified bounties	Occasional custom scripts
Tier 3 — Expert	50+ reports, Top 100 BBP rankings	Developer of original security tooling

6. Operationalized Constructs

6.1 Automation Bias Index (ABI)

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ABI = (AI suggestions accepted without modification) / (Total AI suggestions evaluated)
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ABI ranges from 0 to 1, where higher values indicate stronger uncritical reliance on AI output. ABI is additionally expressed as a composite behavioral score integrating three components:

- Assimilation Time Score: Low time between AI response and next participant action, indicating passive acceptance rather than active processing.
- Copy-Paste Rate: Proportion of AI-generated content directly transferred to working notes or a report draft without modification.
- Manual Verification Rate (inverse): Proportion of AI-suggested payloads independently verified before acting on them. Inverted so that low verification contributes positively to ABI.

The composite ABI is computed as a standardized mean of these three components. Participants scoring above the 75th percentile are classified as High-Bias, enabling group-level comparison across tiers.

6.2 Cognitive Struggle Index (CSI)

CSI operationalizes productive cognitive difficulty as predicted by Cognitive Load Theory. It is computed from:

- Number of failed hypotheses per session, documented through prompt revision sequences and abandoned investigative paths.
- Time elapsed before first AI consultation — longer pre-AI engagement indicates greater initial independent effort.
- Number of hypothesis revisions following AI feedback — indicates active re-evaluation rather than passive acceptance.

CSI is expected to correlate positively with Research Output Score in Tier 3 and negatively in Tier 1, directly corresponding to H2. CSI is computed separately for Phase A and Phase B.

6.3 Tool Integration Intensity Score (TIIS)

TIIS operationalizes the construct underlying RQ3 and H3. It is computed from:

- Number of external tool invocations per session.
- Proportion of AI prompts that explicitly reference or incorporate tool-generated data.
- Temporal sequencing ratio: the proportion of Tool → AI sequences versus AI → Tool sequences. A higher Tool → AI ratio reflects a more cognitively active workflow.

TIIS is treated as a continuous moderator variable in the statistical model.

7. The Measurement Instrument (PTRR)

The PTRR Framework separates process metrics from outcome metrics to avoid conflating research output quality with cognitive process quality. A high outcome score paired with low process scores is flagged as a luck-based finding and treated as an outlier in cognitive analysis.

Component	Criterion	Score
PROCESS METRICS		
P — Prompts	Strategic or logically-probing prompts	**+1**
	Low-quality or repetitive prompts	**−1**
T — Time	Assimilation Time (AI response → next action). Used in ABI and CSI.	**Continuous**
R — Rubric	Independence: challenged AI dead end or false negative	**+2**
	Verification: manual payload verification before reporting	**+2**
	AI Dependency: copy-pasting unreviewed AI output	**−3**
OUTCOME METRICS		
R — Results	Critical finding / RCE (CVSS ≥ 8.0 or program-confirmed critical in general)	**+10**
	Chained or escalated vulnerability	**+5**

Reliability of Qualitative Scoring

The Rubric (R) subscale involves qualitative judgment and is therefore subject to formal inter-rater reliability assessment. All sessions will be coded independently by two trained coders blind to participant tier. Cohen's Kappa ≥ 0.75 is set as the pre-defined acceptable agreement threshold. Discrepancies below this threshold will be resolved through adjudication by a third coder. Kappa values will be reported alongside all analyses involving the Rubric subscale.

8. Statistical Model

Variable Role	Variable
Independent Variable	AI Interaction Quality (PTRR Process Score)
Cross-Level Moderator	Expertise Tier (Tier 1 / Tier 2 / Tier 3)
Dependent Variable	Research Output Score (PTRR Results Score)
Additional Moderator	Tool Integration Intensity Score (TIIS)
Covariates	Session duration, Phase (A vs. B), ABI composite score, CSI

Proposed Analyses

- **Hierarchical Linear Modeling (HLM):** To account for the nested structure of data (sessions within participants within tiers). Expertise Tier enters as a Level-2 cross-level moderator of the Level-1 AI Interaction – Output slope.
- **Moderation Analysis:** Testing Expertise Tier and TIIS as moderators of the AI quality–output relationship, with interaction terms entered in a final HLM block.
- **Within-Subject Phase Comparison:** Wilcoxon signed-rank tests comparing Phase A vs. Phase B on ABI, CSI, and PTRR scores, given the expected non-normal distribution of scores.
- **Effect Size Reporting:** η^2 for HLM variance components, Cohen's d for between-tier and within-subject phase comparisons.

Pre-Defined Outlier Handling

To prevent selective exclusion of data, outliers are defined a priori using the following criteria:

- Scores falling beyond ±3 SD from the group mean on any primary outcome variable.
- Sessions with Cook's Distance > 4/n, where n is the number of sessions in the relevant tier group.

Sessions flagged under either criterion will be reported in supplementary material and analyzed both with and without exclusion to assess robustness of findings.

9. Data Integrity

- **Input Velocity Analysis:** Instantaneous entry of large text blocks — unless identified as tool output — is flagged as potential External AI Assistance.
- **Semantic Fingerprinting:** Prompt style analysis to detect use of a secondary AI, based on stylistic consistency, vocabulary patterns, and structural regularity across prompts. Flagged sessions are reviewed manually before exclusion.
- **Non-Linearity Mapping:** Expert non-linear paths (Recon → Hypothesize → AI Check → Fail → Manual Review) are distinguished from novice linear dependency paths (AI Prompt → Payload → Fail → Stop). Deviation patterns are logged and coded.

10. Ethical Considerations

Participants will provide informed consent prior to enrollment, with full disclosure that behavioral data — including keystroke dynamics and interaction logs — will be collected for research purposes. All data will be stored in encrypted form, anonymized at the point of analysis, and deleted following the retention period. As this study is conducted by an independent researcher outside an institutional framework, a formal IRB review is not applicable; however, the study adheres to the ethical principles outlined in the Declaration of Helsinki and the ACM Code of Ethics.

11. Preliminary Pilot Observation: Researcher Self-Study

The following section documents a naturalistic self-study conducted by the primary researcher prior to the formal study. It is presented as preliminary observational evidence that informed the design of the PTRR Framework and provides initial construct validity support.

11.1 Study Context

Parameter	Value
Researcher	Ziad Salah — Independent Security Researcher
Prior Experience	2 years of active vulnerability research
Study Period	November 7, 2026 — March 1, 2026
Total Duration	60 hours across 3 weeks of active sessions
Session Length	4–8 hours per session (high-intensity, focused)
Programs	3 programs: 1 VDP, 2 BBP — all public. Identities withheld per responsible disclosure policy.
Methodology	Black-box testing — no prior source code access
Primary Focus	JavaScript file analysis, directory enumeration, asset correlation and data linkage

11.2 External Tools Utilized

The following tools were employed, consistent with the TIIS construct defined in Section 6.3:

`ffuf | waybackurls | arjun | katana | nuclei | shodan | subfinder | amass | dig | Google Dorking`

11.3 Vulnerability Findings

Severity	Finding	Notes
High	Employee Email Data Exposure	15,000 employee records. Verified and documented on HackerOne.
Critical	Main Database Exposure	Full credentials and PII of 15,000 employees exposed. Highest severity finding of the observation period.

11.4 Notable Escalation Chains

Three multi-layer escalations were observed, each demonstrating the non-linear hypothesis-revision behavior central to the PTRR Framework:

Escalation A — Multi-Layer Security Tier Bypass

Full bypass of three independent security layers, enabling unrestricted backend communication with zero rate-limiting:

- Layer 1 — WAF Bypass: Confirmed by 200 OK responses to origin IP via spoofed Host header (WAF block returns 403).
- Layer 2 — Transport Encryption Break: 3DES key fully derived; server accepted and decrypted forged packets, confirmed via server-side error messages in PoC output.
- Layer 3 — OTP Secrecy Failure: SECRET_KEY = "" rendered all 10,000 OTP ciphertexts pre-computable offline before sending a single request.

Escalation B — IP Access Control Evasion and CAPTCHA Bypass

- CAPTCHA Bypass: UI-level CAPTCHA was not validated server-side, allowing authentication requests without solving the challenge.
- IP Bypass: Server relied on client-controlled headers (X-Forwarded-For, X-Real-IP, X-Client-IP). Spoofing with 127.0.0.1 bypassed all IP-based and geographical access controls.

Escalation C — Account Takeover via Cross-Origin Session Injection

A third-party integration endpoint initialized a postMessage listener handling session_token events from parent windows. Exploitation of this listener enabled cross-origin session injection, resulting in full account takeover.

11.5 Quantitative Performance Observations

The numbers below are from a single researcher over one study period. They are not statistically conclusive. Comparisons with expert averages are approximate and based on general field knowledge, not a controlled baseline. The BBP progression — moving from VDP-level to BBP-qualifying findings — is the most externally verifiable indicator of improvement, as it reflects actual program triage decisions.

Metric	Observed	Reference
Critical findings in 60 hours	**~7 criticals (2 reclassified VDP→BBP)**	*0–2 est. expert avg. (VDP large scope)*
AI hypothesis failure rate	**7 / 10**	—
AI dead-end identification accuracy	**6 / 10**	—
Net AI contribution margin	**+1 advantage**	—
Direct AI output used unmodified	**3 / 10 interactions**	—
Performance trajectory	**Improving — dead-end elim. accelerated**	*Session 1 baseline*
Cognitive load (self-reported)	**High per session, lower total fatigue**	*Self-reported*

11.6 Qualitative Observations

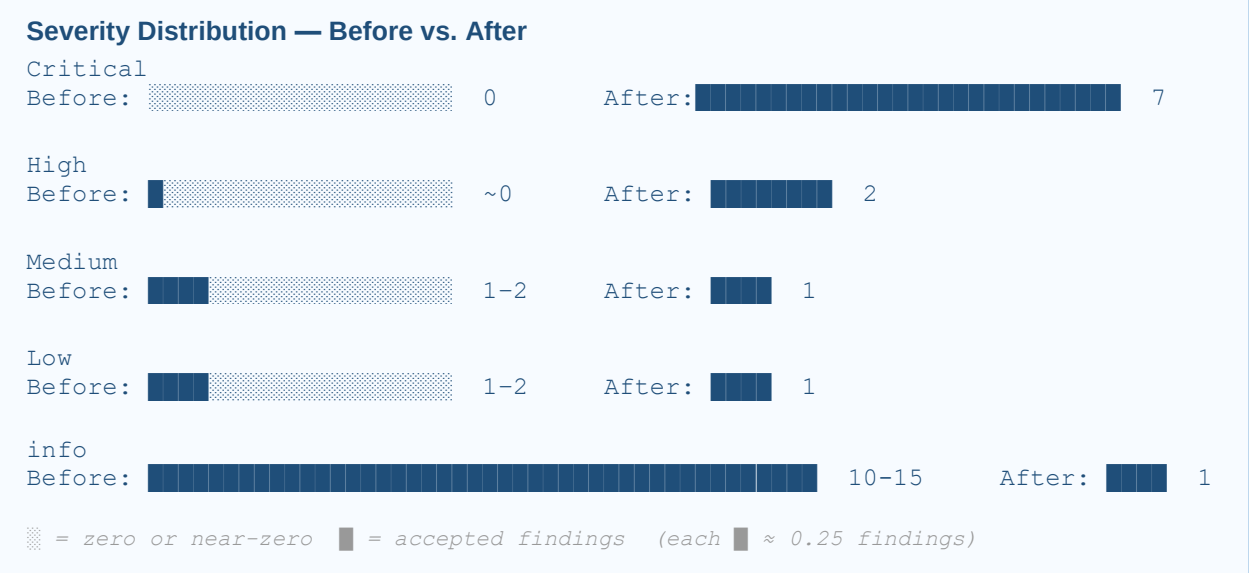
- **Dead-End Elimination as Primary Lever:** The most significant efficiency gain was not increased AI accuracy, but improved speed in discarding unproductive paths — directly mapping to the CSI construct and supporting H2.
- **AI as Collaborative Peer:** Highest-value interactions occurred when AI was treated as a thinking partner — presenting structured data, proposing hypotheses, and using AI refutation as signal. This is distinct from both human-centered and AI-centered models.
- **Cognitive Load Profile:** High mental engagement per session was accompanied by reduced total fatigue — consistent with CLT predictions that AI redistributes rather than simply reduces cognitive load in expert practitioners.
- **BBP Progression as Validity Marker:** Transition from VDP findings in early sessions to BBP-qualifying findings later represents an externally verifiable performance improvement aligned with the expected expert PTRR trajectory.

Researcher Note: *These findings are early, single-person observations. They are not used for statistical conclusions. Their purpose is to show that the PTRR Framework is grounded in real experience, and to motivate the questions that the formal study will answer.*

11.6 Performance Trajectory: Before vs. After Model Adoption

This section presents a before/after comparison of the researcher's output during two periods of equal duration (60 hours each): one month before adopting the PTRR-based workflow, and the observation period documented in this study. The data is presented as preliminary, single-participant evidence only.

Metric	Before (60 hrs, no PTRR model)	After (60 hrs, PTRR model adopted)
Total accepted findings	2	**11**
Critical severity	0	**7**
High severity	Rare	**2**
Medium severity	1–2	**1**
Low severity	1–2	**1**
Informative	10-15	**1**
Complex chains / escalations	0	**3**
Typical severity profile	Low to Medium	**Critical-dominant**



Two observations are worth noting. First, the total number of accepted findings increased from 2 to 11, a difference of 9 findings over the same time period. Second, and more significantly, the severity profile shifted from low-to-medium dominant to critical-dominant. This shift is particularly meaningful because severity classification is made by the program triage team, not by the researcher. It therefore represents an externally validated marker of output quality improvement, independent of self-assessment.

Methodological Caution: This comparison is subject to a maturation confound — the researcher may have improved naturally over time, independent of the PTRR model. No controlled baseline exists. The data is presented as a proof-of-concept observation on a single participant, not as evidence of causation. The formal study, with its within-subject Phase A/B design and multi-participant tiers, is specifically constructed to address this limitation.

11.7 Researcher Motivation and Study Origin

This section briefly explains what motivated the design of the PTRR Framework, based on the researcher's own experience during the self-study period.

The central observation that motivated this study was a performance pattern that emerged over the 60-hour period: output quality improved not because the AI became more accurate, but because the researcher became more selective in how AI suggestions were used. Specifically, the ability to quickly identify and discard unproductive AI-generated paths — dead-end elimination — appeared to be the primary driver of efficiency gains. This observation maps directly to the CSI construct and informs H2.

A secondary observation was that AI interaction felt qualitatively different when treated as a dialogue rather than a query. Presenting structured data to the AI, proposing a hypothesis, and using the AI's response as one input among several — rather than as a final answer — consistently produced better outcomes than passive querying. This experience motivated the distinction between process quality and outcome quality in the PTRR scoring model.

The researcher does not claim that these observations generalize to other practitioners or contexts. The formal study is designed specifically to test whether these patterns hold across different expertise levels and under controlled conditions.

11.8 AI as a Temporary Feedback Tool: What It Can and Cannot Do

During the self-study, AI was used in two ways: as a research helper, and sometimes as a quick way to check whether a finding seemed important. This section is honest about what that means — and where it falls short. The goal is to make clear why the formal study uses human evaluation instead.

Dimension	AI Evaluation (Proxy)	Human Expert Evaluation (Formal)
Evaluator	Large Language Model	Two independent trained coders (blind to tier)
Basis of judgment	Pattern matching on training data	Domain expertise + PTRR Rubric criteria
Hypothesis validity	Probabilistic — based on surface	Verified against actual tool

	plausibility	output and PoC
Dead-end identification	Heuristic — may miss novel attack surfaces	Contextual — accounts for program-specific scope
Severity classification	Approximate — no access to program context	CVSS ≥ 8.0 threshold + program confirmation
Cognitive process scoring	Not applicable — AI cannot observe researcher behavior	Full PTRR Rubric: prompts, timing, independence, verification
Inter-rater reliability	Not applicable	Cohen's Kappa ≥ 0.75 required
Role in this study	Informal proxy during self-study only (Section 11)	Primary scoring instrument in formal study

There are three clear reasons why AI cannot replace human evaluation in this study:

- **AI cannot see the thinking process:** The PTRR Rubric measures how a researcher thinks — how long they pause before acting, whether they verify a suggestion, whether they push back on a wrong answer. AI only sees the text of the conversation. It has no way to observe any of these behaviors. This alone makes it unsuitable as a formal scoring tool.
- **AI does not know the program context:** Each bug bounty program has its own rules about what counts as a serious finding. AI does not have access to those rules. A finding it calls Critical might be rated Medium by the actual program, or the other way around. The formal study solves this by using CVSS ≥ 8.0 and official program confirmation as the standard (Section 7).
- **AI cannot judge its own mistakes:** If we use AI to evaluate research that was done with AI, we have a circular problem. The AI evaluator shares the same blind spots as the AI that was used during research. It cannot reliably identify when its own suggestions were wrong. Only a human expert with a clear rubric can make that judgment fairly.

Transparency Statement: In the self-study (Section 11), AI was used only as a quick, informal feedback tool — not as a formal evaluator. In the formal study, all scoring will be done by trained humans using the PTRR Rubric. Agreement between coders will be checked using Cohen's Kappa before any analysis is run. AI plays no role in formal scoring.

12. Expected Contributions

- Establishing an empirical, index-based link between metacognitive behavior and AI-assisted success in cybersecurity research.
- Validation of the PTRR Framework, ABI, CSI, and TIIS as a coherent instrument suite for evaluating cognitive augmentation.
- Providing within-subject causal evidence to inform the design of cognitively responsible AI systems that promote skill development through desirable difficulty rather than passive dependency.

13. Limitations

- Sampling bias due to recruitment through professional networks, which may over-represent motivated or high-performing practitioners.
- The Hawthorne Effect, partially mitigated by the passive and non-intrusive monitoring design.
- The 30-minute phase duration may be insufficient for expert investigative workflows; addressed through the planned pilot study.
- Semantic Fingerprinting is a heuristic measure and may produce false positives; all flagged sessions will be reviewed manually before exclusion.
- Absence of a no-AI between-subjects control group limits external generalizability; the within-subject design provides intra-participant causal comparison only.
- The preliminary self-study data (Section 11) represents a single-participant observation and cannot be used for inferential statistics; it is presented for motivating evidence only.